VISUAL GUIDE

Microsoft Agent Framework & Foundry Agent Service

What they are, how they work, and how they fit together

BUILD

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Agent Framework

Open-source SDK you write code with

deploy as a

hosted agent

RUN

Foundry Agent Service

Managed Azure platform that hosts and runs agents

1. **Agent Framework** is the toolkit for designing agents and multi-agent workflows in .NET and Python.
2. **Foundry Agent Service** is Microsoft's fully managed service for building, deploying and scaling agents.
3. **Together** you write the logic in the framework and let Foundry host, secure and monitor it.

Prepared November 2026 | Based on Microsoft documentation and announcements

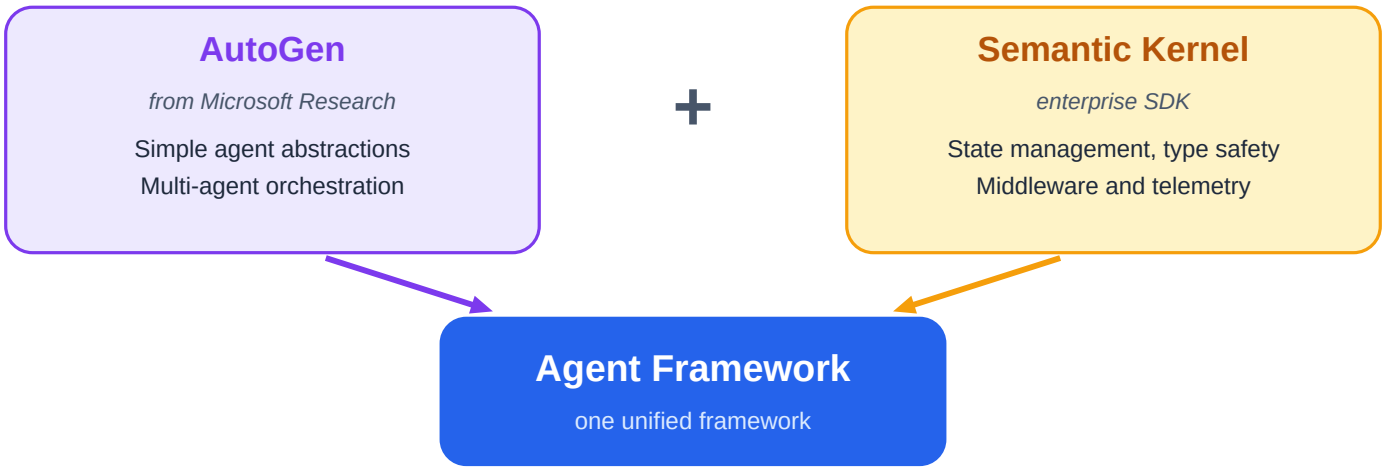
Microsoft Agent Framework

The open-source SDK for building agents and multi-agent workflows

What it is

Microsoft Agent Framework is an **open-source development kit** for building AI agents and multi-agent workflows. It is the **direct successor to Semantic Kernel and AutoGen**, built by the same teams, and brings the best of both into one framework so developers no longer have to choose between research innovation and production stability.

Where it came from



Plus new capabilities neither predecessor had:

Graph-based workflows | Checkpointing | Human-in-the-loop

Languages: .NET | Python | Go (preview)

Two pillars

Agents

- Use an LLM to understand input and decide what to do
- Call tools, functions and MCP servers
- Keep conversation history in threads, so chats can be long-running and resumable
- Work across model providers

Workflows

- Graph-based orchestration of many agents and functions for complex, multi-step tasks
- Checkpoints so long processes can pause and resume
- Human-in-the-loop steps for approvals
- Explicit, predictable flow control

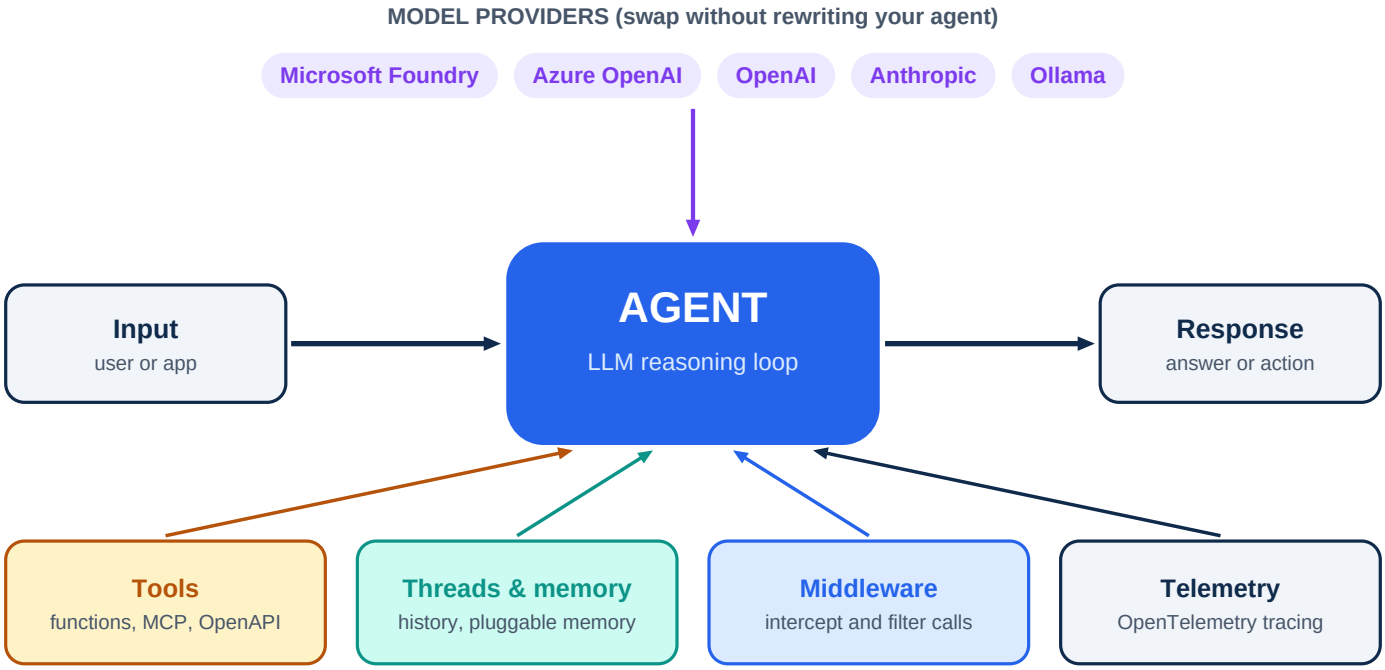
Who it is for

Developers who want **one unified framework** instead of piecing together several agent libraries, especially for long-running processes, multi-agent systems and human-in-the-loop scenarios. The same code can go from prototype to production, across providers and deployment targets.

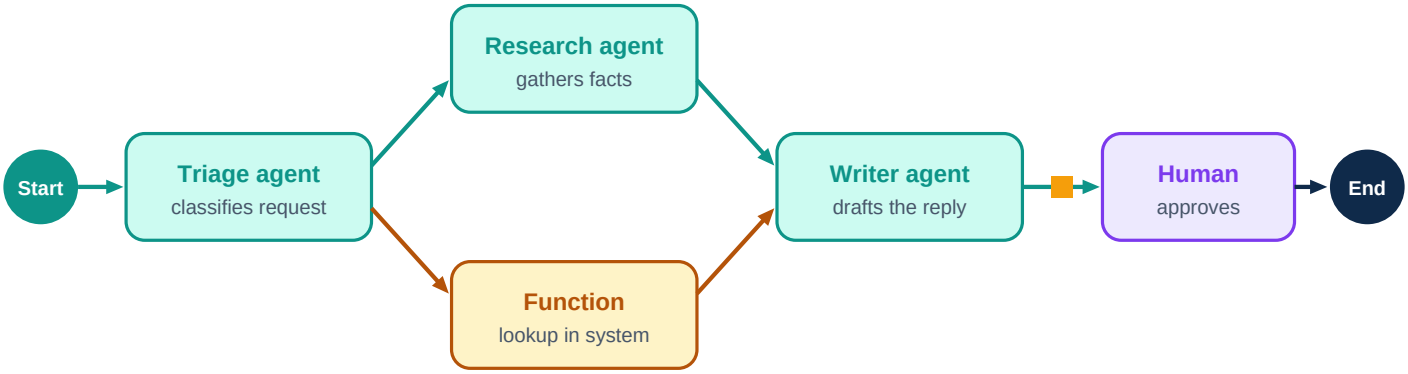
Inside the Framework

How a single agent works, and how workflows connect many of them

Anatomy of an agent



Example workflow (graph-based orchestration)



■ = checkpoint: state is saved, so the workflow can pause (for example, waiting on the human) and resume later

Built-in features

Open standards
MCP, Agent-to-Agent (A2A) and OpenAPI support

Enterprise connectors
Built-in connectors to enterprise systems

Declarative agents
Define prompts and agents in YAML or JSON

Observability
OpenTelemetry tracing out of the box

Pause and resume
Long-running and human-in-the-loop flows

Ecosystem
Integrates with Foundry, MCP, A2A and M365 Copilot agents

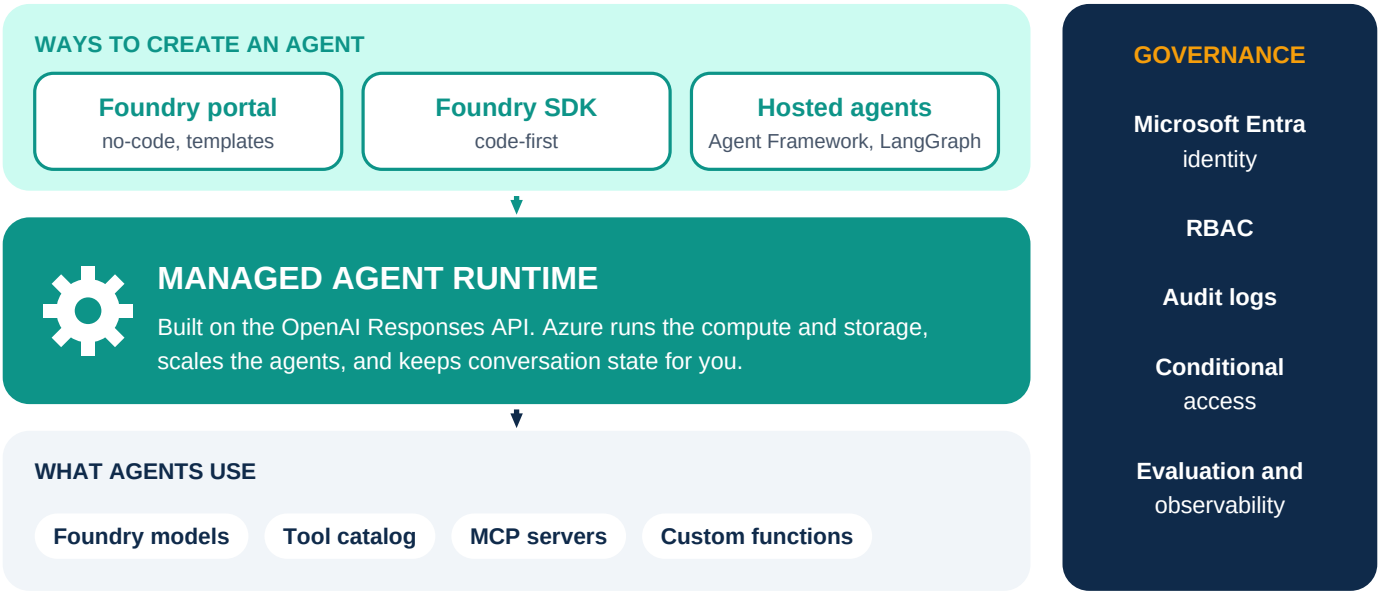
Microsoft Foundry Agent Service

The fully managed Azure platform for running agents in production

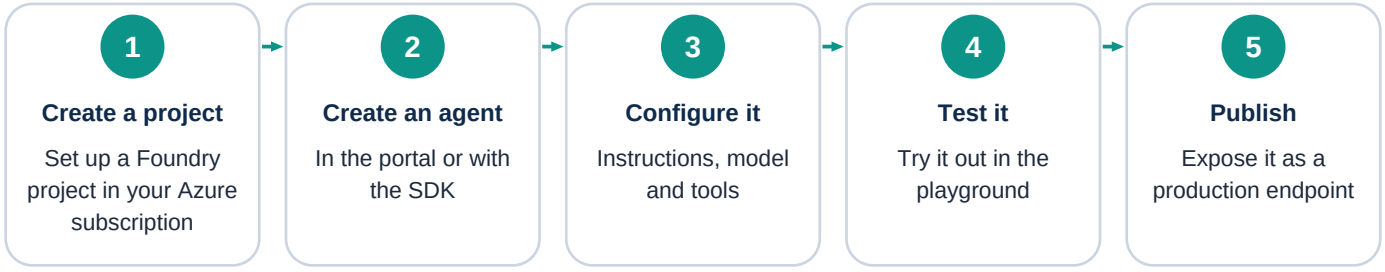
What it is

Foundry Agent Service is a **fully managed service** for securely building, deploying and scaling AI agents **without managing the underlying compute and storage**. You can create agents in the Foundry portal or in your own code with fewer than 50 lines. Announced at Ignite 2025 as a unified agent platform built on the **OpenAI Responses API**. It was previously known as Azure AI Agent Service.

How the platform is layered



Getting started in five steps



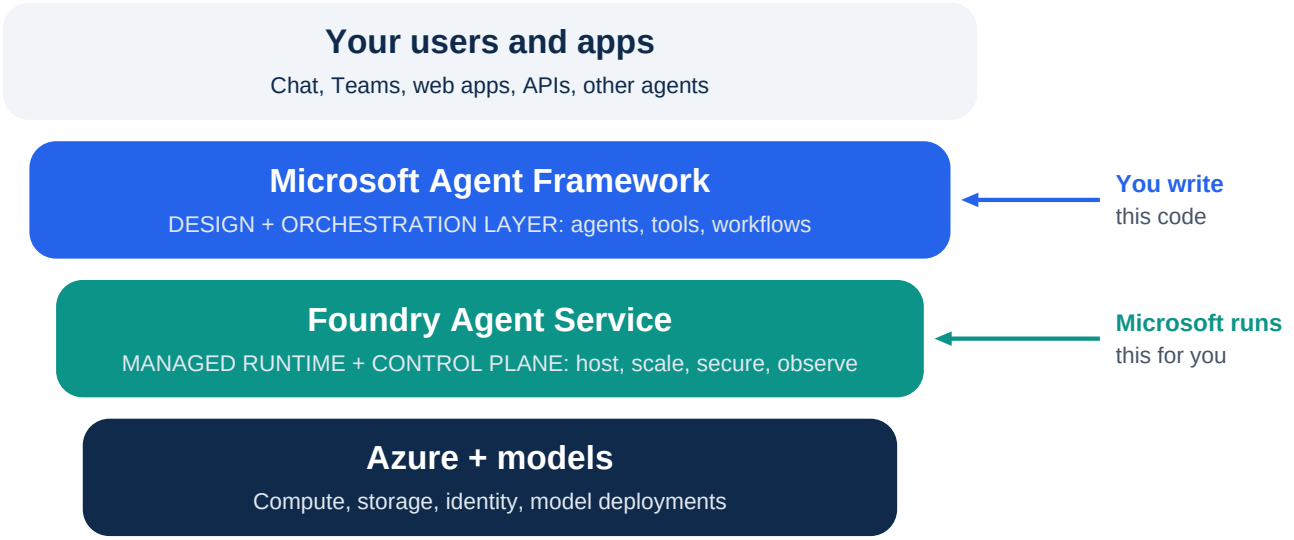
Who it is for

Enterprises and development teams that need **secure, governed and scalable** agents to automate business workflows, and want Azure to handle hosting, identity and monitoring. Pricing is pay-as-you-go on Azure.

How They Fit Together

Complementary, not competing: one designs the agent, the other operates it

The stack



Three ways to use them

A — Framework only

Agent Framework

Write agents in code and run them on your own infrastructure with any supported model provider.

B — Agent Service only

Foundry Agent Service

Build agents in the Foundry portal or with the Foundry SDK, with no framework code needed.

C — Both together

Framework → Foundry

Write the logic in Agent Framework, then deploy it to Foundry as a hosted agent for hosting, security and monitoring.

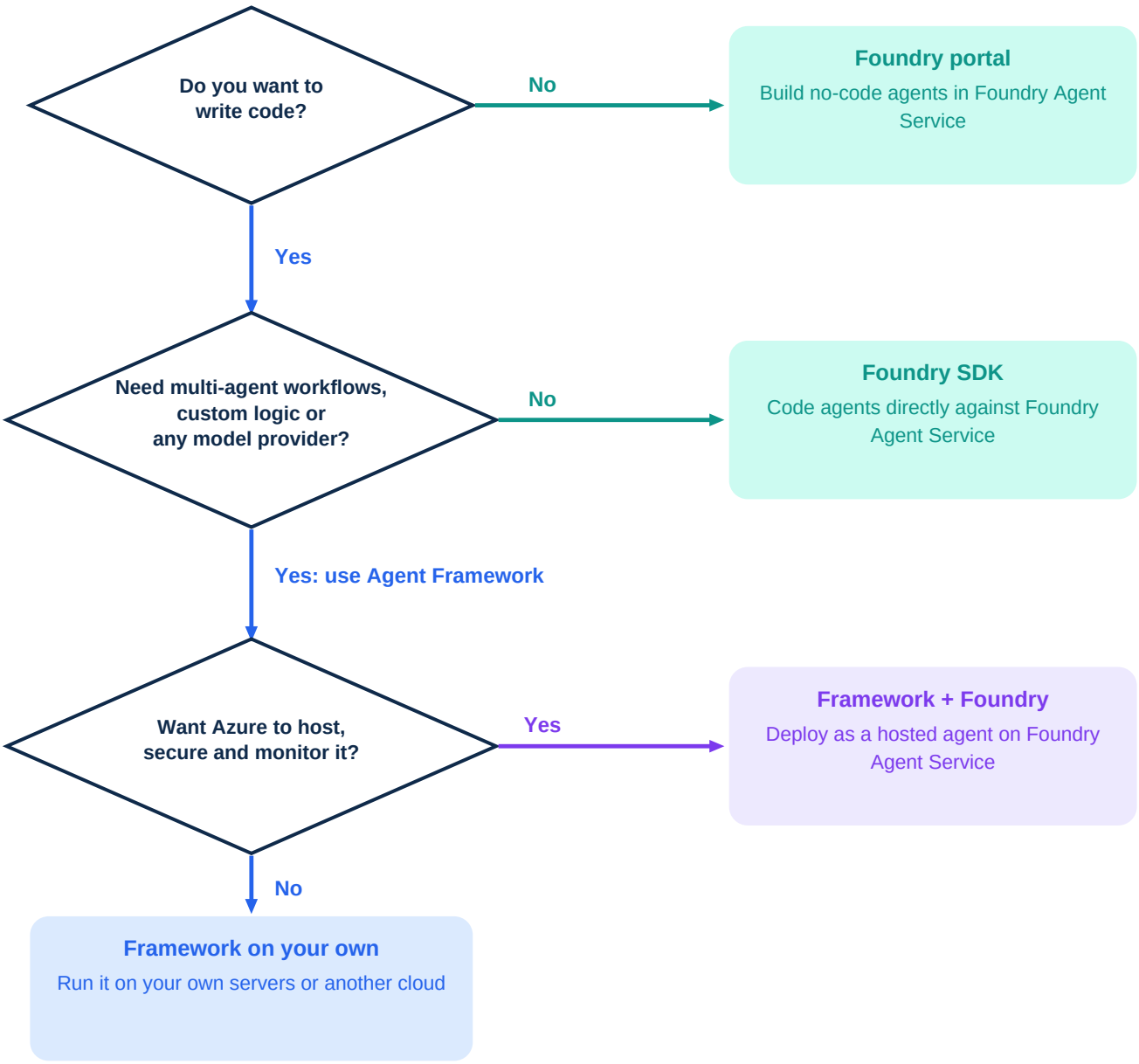
Side by side

	Agent Framework	Foundry Agent Service
What it is	Open-source SDK and runtime library	Fully managed Azure platform
Main job	Design agents and orchestrate workflows	Host, scale, secure and observe agents
Where it runs	Anywhere: laptop, your servers, any cloud	Microsoft Foundry on Azure
Models	Many providers (Foundry, OpenAI, Anthropic...)	Models deployed in Foundry
How you use it	Code in .NET, Python (Go in preview)	Portal (no-code), SDK, or hosted agents
Cost	Free and open source	Pay-as-you-go Azure pricing

Which Should You Use?

A quick decision guide, plus how to get started

Decision guide



Install Agent Framework

Python

```
$ pip install agent-framework
```

.NET

```
$ dotnet add package Microsoft.Agents.AI
```

Learn more

- **Agent Framework docs:** learn.microsoft.com/agent-framework
- **Agent Framework source code:** github.com/microsoft/agent-framework
- **Foundry Agent Service docs:** learn.microsoft.com/azure/ai-foundry/agents/overview

Note: some features (such as the Go SDK and hosted agents) were in preview at the time of writing and may have changed.